Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: SQL Development, Advanced SQL (Joins, CTEs, Window Functions, Stored Procedures), Query Optimization & Execution Plan Analysis, Indexing & Partitioning, ETL/ELT Pipeline Design, Data Warehousing (Star/Snowflake Schemas, SCD Type 1/2, CDC), Incremental Load & Batch Processing, Data Validation & Reconciliation, Performance Tuning, Predictive Machine Learning, Agentic AI Workflows, Time-Series Forecasting (SARIMA), MLOps.

Programming Languages & Tools: Azure Data Factory (ADF), Azure Databricks, Azure Synapse, SQL Server, Snowflake, AWS (S3, Redshift), Apache Spark / PySpark, Oracle, PostgreSQL, Power BI (DAX), Tableau, Collibra, Git, KIRO IDE, Python.

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Senior Data & SQL/ETL Pipeline Engineer — **May 2022 – Present**

- **Architected high-throughput cloud ETL pipelines** and dimensional models (Star/Snowflake schemas, SCD Type 1/2) using Azure (ADF, Databricks, Synapse, SQL Server), Snowflake, and PySpark, ensuring 99.9% data availability SLAs for enterprise analytics.
- **Engineered autonomous Agentic AI workflows** utilizing LLM prompt orchestration and KIRO IDE to automate schema discovery, data profiling, regex classification, and complex code migration, reducing manual delivery turnaround by 40-50%.
- **Optimized complex SQL queries and ETL performance** by conducting rigorous execution plan analysis, implementing advanced indexing strategies, partitioning, and tuning stored procedures, joins, CTEs, and window functions across large-scale relational databases.
- **Implemented source-to-target reconciliation firewalls** and automated data quality governance frameworks integrated with Collibra data lineage, reducing operational discrepancy resolution turnaround by over 80%.
- **Deployed autonomous ML anomaly detection** and incremental batch processing pipelines identifying schema variations, distribution drift, and ETL discrepancies with 95% precision, safeguarding downstream data warehouse integrity.

Data Scientist & Machine Learning Specialist — **May 2022 – Present**

- **Built predictive machine learning models** using XGBoost, Random Forests, and Scikit-Learn classification algorithms to analyze customer behavior and deliver actionable signals that improved service reliability by 20%.
- **Architected advanced time-series forecasting models** (SARIMA) to predict truck rolls and field support demand, optimizing workforce capacity allocation and reducing operational response times by 20%.
- **Engineered automated testing and evaluation frameworks** for conversational AI platforms (SpectrumIQ), cutting manual QA turnaround by 50% and accelerating production release cycles for enterprise data solutions.

Business Intelligence & Decision Analytics Specialist — **May 2022 – Present**

- **Designed semantic models and Key Operating Metric** (KOM) dashboards in Power BI (DAX) and Tableau, translating complex transactional telemetry and data warehouse outputs into executive corporate strategy.
- **Executed comprehensive data validation and profiling** across massive data stores, bridging complex business requirements into efficient ETL solutions and robust data integration workflows.
- **Partnered with cross-functional teams** using Agile methodologies and Git version control to deliver scalable data warehouse solutions and high-performance reporting architectures.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.